

MTE 203 Week 3 Tutorial

Trim Ch 11, S. 11.10, 11.13



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Problem 1: (Ch11.10, Prob 5)

Find the parametric and vector representations for the curve given by the intersection of the surfaces

$$z + 2x^2 = 1$$

and

$$y = z$$

directed so that x decreases along the curve. Sketch the curve indicating its direction.

Solution

Problem 1 - Solution

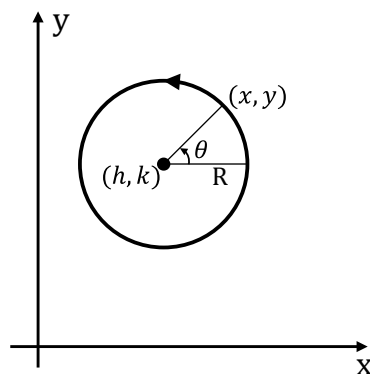
Problem 1 - Solution

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Problem 1 - Solution

Problem 2 [S.11.13, P.16]

A particle travels counterclockwise around the circle $(x - h)^2 + (y - k)^2 = R^2$ drawn in the figure below:



- Show that the speed of the particle at any time is $|\vec{v}| = \omega R$ where $\omega = \frac{d\theta}{dt}$ is called the angular speed of the particle.
- Find the unit tangent vector to the curve at the point $P_0 = \left(\frac{R}{\sqrt{2}} + h, \frac{R}{\sqrt{2}} + k\right)$.
- Write a line equation for a tangent line passing through the point $P_0 = \left(\frac{R}{\sqrt{2}} + h, \frac{R}{\sqrt{2}} + k\right)$.



Problem 2 - Solution

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Thank You